

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech. Examination (CE)****May-June 2018****CE745 Intelligent Networks****Date: 03.05.2018, Thursday****Time: 01:30 pm To 04:30 pm****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I**Q-1 Answer the following questions. [11]**

- [A] How does the hop-count-limit reduce RIP's problems? [2]
- [B] Why hidden station problem exists in wireless network but does not exist in wireline network? [2]
- [C] What is ICMP? What kind of errors are resolved by ICMP? [2]
- [D] What is connection oriented and connection less services? [2]
- [E] What is classfull and classless addressing? Where it is used? [3]

Q-2 Answer the following questions. (Attempt any THREE) [12]

- [A] What is method to recognize a class in IPv4 address? Also, write down total number of host and network available in each class.
- [B] Network layer and data link layer uses their own addressing scheme. What protocol does mapping between different addresses? How does it achieve?
- [C] Differentiate between circuit and packet switching network.
- [D] What are the limitations of current networking technologies so that there is need of software-defined network?

Q-3 Answer the following questions. (Attempt any TWO) [12]

- [A] By taking suitable diagram, explain NDN entities and its functionality.
- [B] By taking suitable example, Explain the working of RIP routing table generation.
- [C] Draw and explain fields of Ethernet frame format. Also, explain functionality of each field.

SECTION-II**Q-4 Answer the following questions. [11]**

- [A] Why is UDP protocol used in live video streaming although it is connection less protocol? [2]
- [B] How does transport layer achieve process-to-process communication? [2]
- [C] Why three-way handshake is required during connection establishment in TCP? [2]
- [D] Why two half close is used in connection termination instead of three-way termination? [2]
- [E] How is checksum calculated at transport layer? [3]

Q-5 Answer the following questions. (Attempt any THREE) [12]

- [A] Draw and Explain TCP State Transition Diagram.
- [B] Named Data Network does not use IP address for host identification. How NDN does propagate request into network?
- [C] What kind of problem arises when either sender or receiver sends/receives data slowly? Write down the solutions of it.
- [D] What is importance of southbound and northbound interface in SDN? What protocols are managed by each interface?

Q-6 Answer the following questions. (Attempt any Two)

[12]

[A] Draw and explain SDN Application Plane Architecture.

[B] Explain Additive increase and multiplicative decrease method of transmission control protocol.

[C] List and explain timers used by TCP to control state transitions.